CUTLIST — finding the promos

Problem 1. A tool that reads a microdrama screenplay and returns ranked 20–60s promo clips with exact cut points. Run on all 50 episodes of *I Married a Broken Soldier. He was a Duke*. Repo: **github.com/Mr-Bathwal/Kuku_Assesment** · Prototype: `web/index.html`, opens off the filesystem, no server · Raw outputs: `out/` · Run steps: `README.md`

Why this problem

Marketing was **₹1,105 crore of Kuku's FY26 spend — 74.5% of revenue and 77.7% of total expenditure, about ₹3 crore a day.**[1] Microdrama apps run roughly **5,770 deduplicated ad creatives a month each**, and one hit show has run **98,000**.[2] The ad-intelligence teardown of the biggest spender is blunt about what wins: *"not long videos about its overall plot, but under-1-minute clips of exciting moments."*[2] That is precisely this output. Promo selection is a hand step sitting on top of the largest cost line in the company — and Kuku ships **200–250 shows a month, targeting 1,000**, with dedicated promo editors *per language*.[3] Problems 2 and 3 are real, but neither touches a ₹3-crore-a-day lever.

What I assumed, where the brief was silent

- **A cut may only land on a beat boundary** written in the screenplay — never mid-line. This is the central design constraint, not a detail.
- **Promos are mostly one continuous stretch.** Microdrama episodes are written as one location, one beat, so a promo is a *trim of a scene*, not an assembly.[4] I still generate stitched promos — the brief allows them — but conservatively: one hook, one cliffhanger, nothing between.
- **Timecodes are MM:SS.cc, 30fps** for EDL export.
- **No video exists**, only screenplays, so everything is judged from text.
- **"Best" means best for cold acquisition** — a stranger mid-scroll — not best scene.

How it works

Taste is a model's job; arithmetic is not. Where a clip starts and stops, how much dialogue is in it, how many names a stranger must track — countable, free, instant, impossible to hallucinate. The model is spent only on what counting cannot judge.

```
parse → signals → candidates → verify → judge → export
```

Parse turns 3,727 blocks into typed, timed beats. Dialogue keeps its actor direction (`Cold, Commanding`) — 620 distinct tags, and the richest signal in the corpus. **Signals** score each beat from hand-built lexicons (shock, reveal, threat, emotional heat, questions, cliffhanger sound *and* shot grammar, resolution, context debt), each recording which beat fired it. **Candidates** generates every window with real beat boundaries at both ends, 20–60s, scored on five axes — **hook** (is there something in the first 3 seconds), **cliff** (does it break before the answer), **intensity**, **standalone** (can a stranger follow it), **shape** — then suppresses near-duplicates so the list is ten promos, not ten crops of one.

Standalone is the axis most tools skip, and the reason good scenes make bad promos: it counts speaking characters (two in confrontation is ideal, five is not), phrases pointing off-screen, and *series baggage* — a character first met in episode 38 carries 37 episodes of history a cold viewer does not have.

The model never receives a timestamp field. It is shown a window the parser already chose and asked only to score hook/cliff/standalone and write the *why*. A hallucinated cut point is structurally unable to exist. For scale: across this show there are **3,419 real beat starts among 61,612 expressible timestamps — a model free-handing an in-point would be wrong 94.4% of the time.** Then `verify` re-checks every cut against the script anyway, before anything is written.

Output is JSON, a CSV to schedule from, a markdown cut sheet, and a **CMX3600 EDL per promo** that imports straight into Premiere or Resolve — the cuts arrive on the timeline, already named and in order.

How I know it's good — and where it isn't

I hand-cut the best promo in **six episodes, chosen and labelled before running the tool on them** (`cutlist/evaluate.py`), then measured:

	mean IoU	gold covered	hits @ IoU ≥ 0.5
cutlist	**0.380**	0.425	2/6
last-40-seconds	0.349	0.500	**3/6**
densest-dialogue	0.090	0.133	0/6
random	0.015	0.026	0/6

It beats the trivial baseline, and not by much. `last-40s` is strong because microdramas are *written* to end on a cliffhanger, and it still covers more of my hand-cut windows than I do. What it cannot do is rank, cut shorter than 40s, find a second promo in an episode, stitch across episodes, or say why — it returns 50 undifferentiated windows. CUTLIST returns 156 verified, ranked, diverse candidates with reasons, in 1.6s.

It runs on a show it has never seen. `data/sample/` is three episodes of an original thriller I wrote for the purpose — different genre, different vocabulary, nothing tuned for it. 104 beats, 0.03s, 11 verified candidates; the top pick opens on a phone slammed onto a table

and cuts out on "*What was her name?*" Writing that test found real overfitting: `"duke": 0.35` sat in the reveal lexicon, learned from this pack's period setting. Removing it moved the gold score by **0.000** — it was never earning its place.

Measurement found three real bugs, each worth more than the tuning: the cliffhanger scorer read only dialogue and sound, so it scored **0.00 on all six hand-cut out-points** — in a screenplay the last beat before a cut is almost always a camera direction, and the shot holding on a reaction *is* the cliffhanger; every hyphenated lexicon term (`close-up`, `push-in`) silently matched nothing; and sound was the only beat kind never scored for shock, so episode 18's falling roof registered as ambience. Fixing those took the score from **0.227 → 0.380**.

Where it picks badly.

- *Lexicon collisions.* "He **strikes** flint and steel over dry kindling" scored as hard as a slap and put a man lighting a campfire near the top. Damped by checking the object the verb lands on — a partial fix, not a real one.
- *Ablation is inconclusive on two axes.* Dropping **hook** or **standalone** *improves* agreement by +0.118 — but it changes only 3 of 6 episodes, two up and one down. At n=6 that is noise. I kept both: the 3-second hook is the primary variable in every industry account of this format,[2] and I'd rather be wrong on six labels than ship a promo finder that ignores the opening.
- *Six labels is a small gold set*, labelled by me, who also wrote the scorer. Real validation is editor A/B against posted performance.
- *It cannot see.* A silent held look can be the best promo in an episode and will score near zero.
- *Spoilers are unguarded.* Nothing stops it proposing the episode-45 reveal as promo #1.
- *Stitched promos are the weakest output* — the research says the form runs on continuous trims,[4] and mine are plausible rather than proven.

Weights are stable: **98% of picks unchanged** when every weight is jittered ±20%.

Scaling to every show Kuku makes

The rules path is **1.6s per 50-episode show and costs nothing** — 225 shows a month is **6 minutes of compute**; the 1,000-show target is 27 minutes. The model pass is the only variable cost and it is deliberately small: only the top 25 candidates are reviewed, at ~852 input + 120 output tokens each — **21K input tokens per show**, ~4.8M a month at today's rate, 21M at the target. Cents, not rupees, per show. Provider is one environment variable (`CUTLIST_PROVIDER`), so a stronger model is a config change — and that path is tested over real HTTP against a local OpenAI-compatible server (`tests/test_model_swap.py`), including a model that volunteers timestamps it was never asked for. They are discarded and the cut points do not move. I have not run it against a paid key.

The human stays at the end, not the middle. The prototype (`web/index.html`, no server, no build) opens on the corpus — fifty episodes as a scroll-driven reel, each frame the episode's own pressure curve — and cuts into a keyboard-driven review desk: approve, reject, nudge an in-point by one beat, export the decisions. Every number on it is read from the run, and because there is no video of this show there are no stills on it either. The editor stops hunting and starts judging. *Assumption:* if promo selection is ~2 hours per show, 225 shows is ~450 editor-hours a month; I could not source a real figure for that step, so treat the saving as directional.

AI tools, and where they got it wrong

Built with **Claude** (Claude Code) for the code, three research subagents for the microdrama, Kuku and UI-reference briefs, and my own reading of the screenplays for the gold labels. It was wrong in a way worth reporting: the first scoring function summed all signal axes for every beat, so a line *screamed* in terror scored **0.22** — below a murmured question — because physical-shock terms only fire on action beats and dialogue could never reach them. It looked completely reasonable and was badly wrong, and only printing the top-scoring lines and not recognising them caught it. Every number above is from a run I can reproduce, which is the only reason I trust any of it.

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[1] RoC filings via Medianama/Entrackr, FY26. [2] SocialPeta × AMO Pictures, *Insight into Global Micro Drama App Marketing*, 2026. [3] Hollywood Reporter India; Kuku job postings. [4] Microdrama writing/editing practice; see `docs/RESEARCH.md`.